Lesson notes · for you

Shortest and best: unit check

Optimization · Level 1 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Pack the truck
- The shortest way
- Puzzles with rules
- Fair turns without clashes

Activity: Activity: Puzzle stations

1. Truck (5 min): With the six loads from lesson 1, find two loads that fit in 1,000 kg and add up their points. Which is best?
2. Map (5 min): Copy the village map and use the closest-first trick to find the shortest way from home to school.
3. Sudoku (5 min): Swap a 4 × 4 sudoku with a partner and solve it by crossing out. Count your guesses.
4. Turns (5 min): Make a fair rota for a class job so everyone has one turn before anyone has two.

Do the stations alone or with a partner, on paper. No device needed.

Discussion

- Which idea from this unit will you use first?
- When did a quick way or a quick look fool you?
- How would you explain fair turns to a younger child?

Check yourself

- My favourite idea from this unit + where I'll use it.

Try it this week
Pick your favourite idea from this unit (the best load, the closest-first trick, crossing out, or fair turns) and use it this week at school or at home. Then explain it to someone younger.

Worksheet

Shortest and best: unit check

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Name _____ Class _____ Date _____

1 **The truck can carry 1,000 kg. Which load is the best?**

- ◯ Solar panels, water filter and chalk: 1,000 kg, 60 points
- ◯ Desks and books: 1,000 kg, 65 points
- ◯ Solar panels and desks: 1,100 kg, 75 points

2 **The best load is the one with the most points, even if it is too heavy.**

- ◯ True
- ◯ False

3 **Which of these are rules: things you must keep? Choose all that apply.**

- ☐ No more than 1,000 kg on the truck
- ☐ No number twice in a sudoku row
- ☐ The most points
- ☐ Two joined tests can't share a time
- ☐ The shortest way

4 **Put the closest-first trick in order.**

- ☐ Stop when the school is settled
- ☐ Give its neighbours smaller numbers if you can
- ☐ Settle the waiting village with the smallest number
- ☐ Give home the number 0

5 **The bridge between Kesi and Mina is flooded. What is the shortest way to school now?**

- ◯ Through Rubi and Sana: 14 km
- ◯ Through Bora: 15 km
- ◯ Through Kesi and Mina: 13 km